Escaping False Valleys

Royal Statistical Society & *Significance* Magazine

Statistical Excellence Award for Early Career Writing 2026

Written & Illustrated by: Saf Flatters | Bachelor of Science (Data Science) | Perth, Western Australia

Curtin University

“She had stopped trusting the mountain about an hour ago.
After the storm, the trail had disappeared and the light dropped faster than expected.
Her phone battery was nearly dead, not that it mattered much because she had no…



Escaping False Valleys

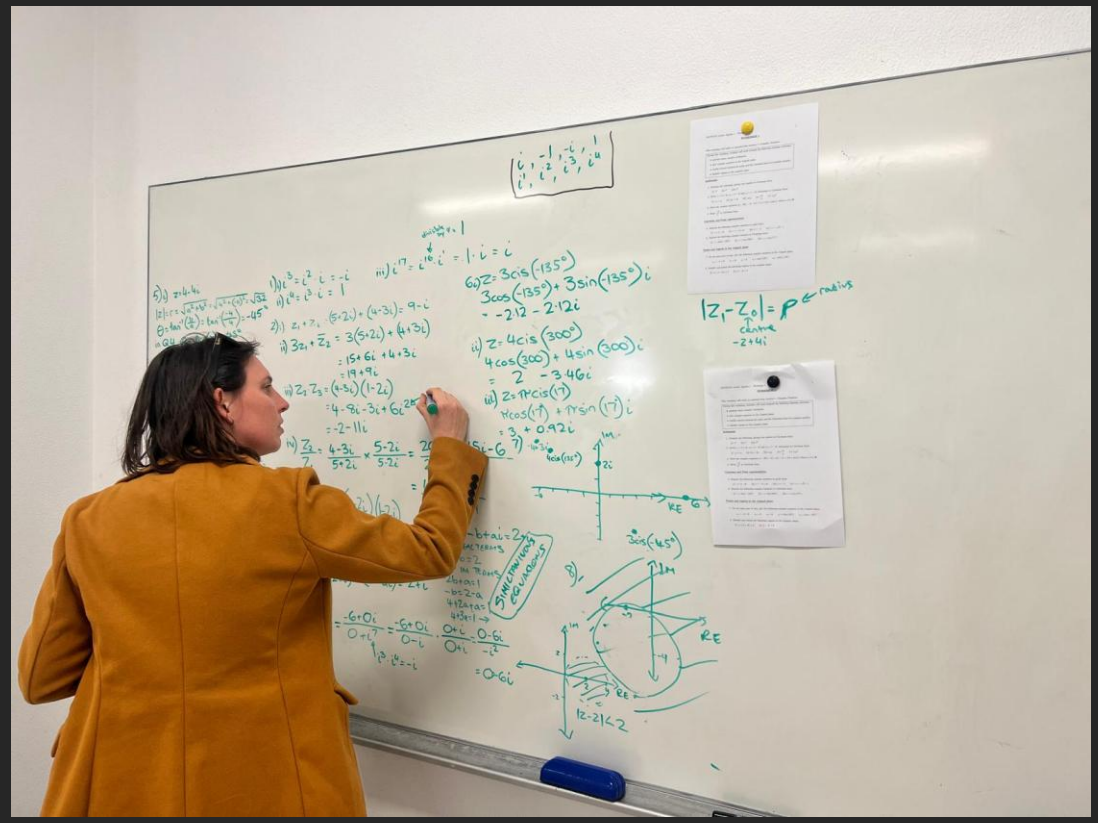
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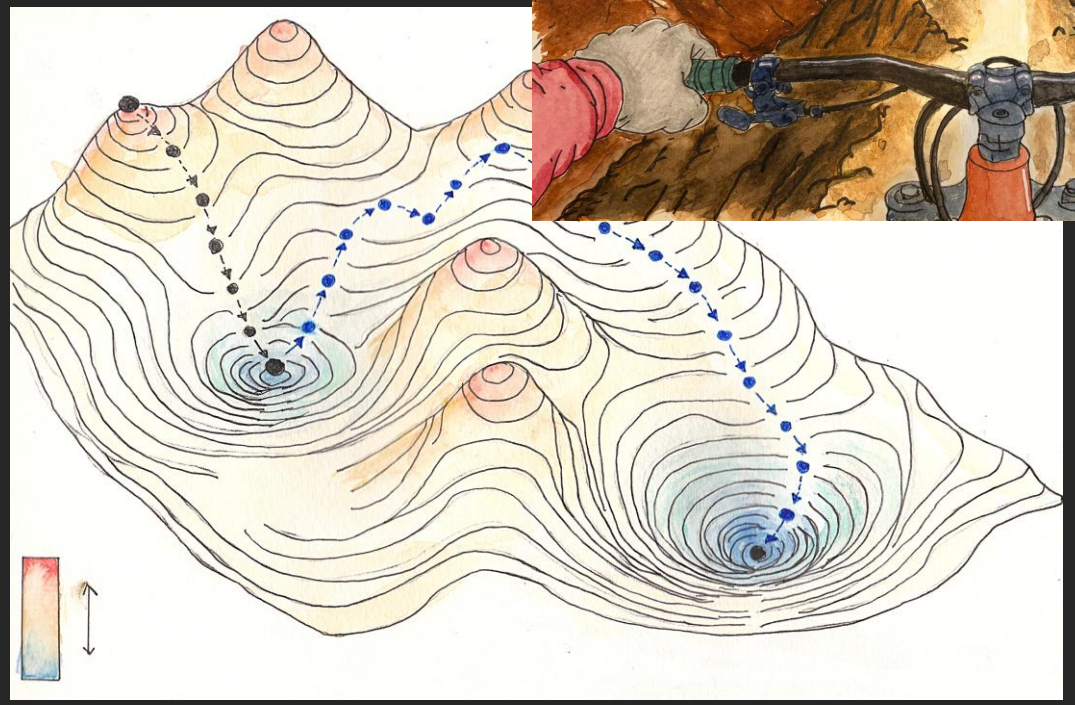
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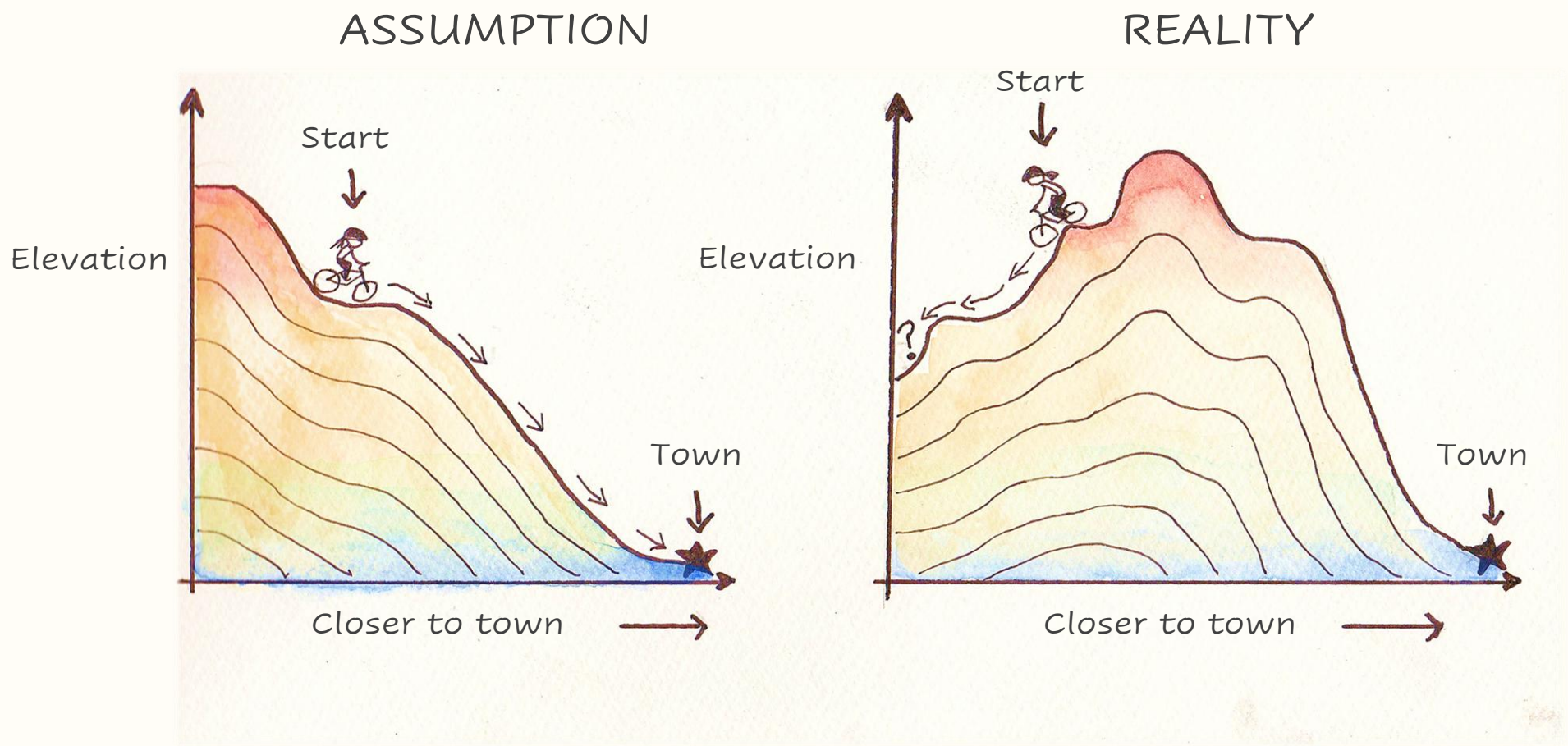








“At every decision point, she had greedily chosen the steepest immediate drop, assuming that what looked best locally would lead to the best outcome globally. Sometimes that works. Sometimes the landscape is simple and predictable…



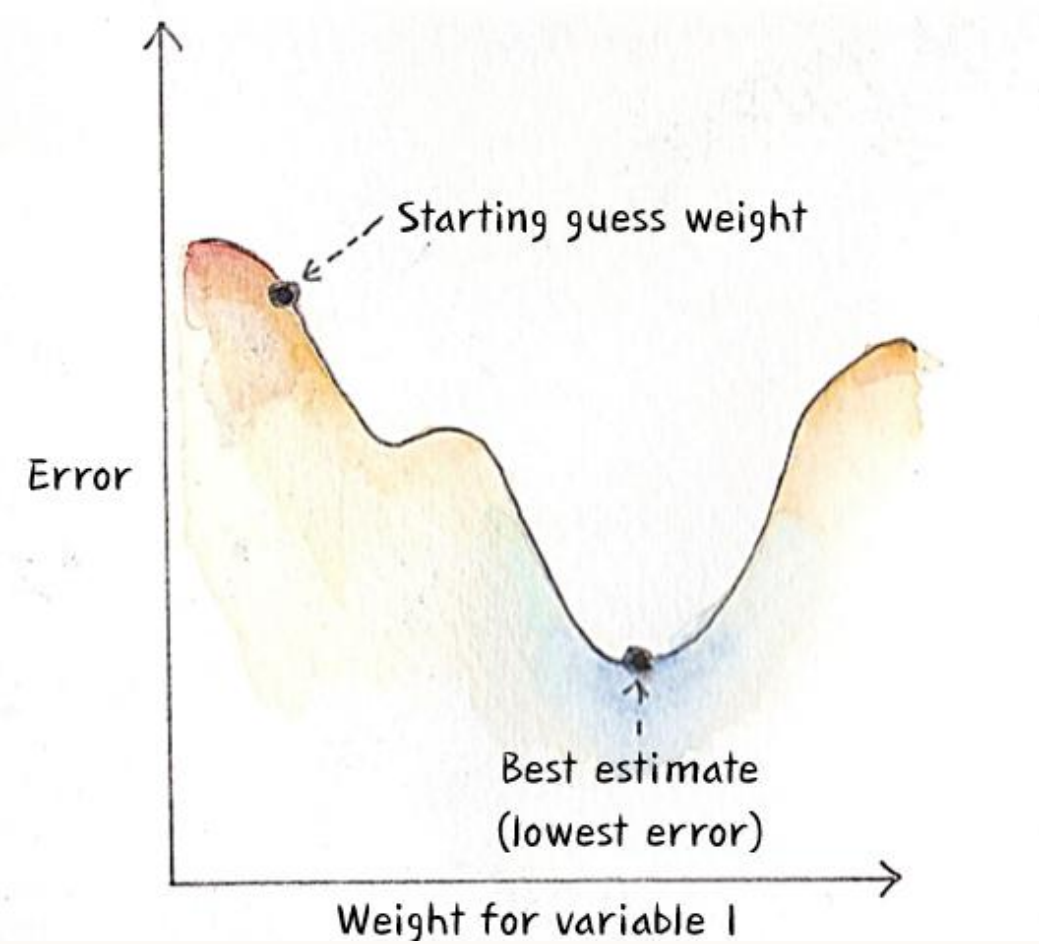
- Global Optimisation
- Stochasticity
- Bayesian Reasoning

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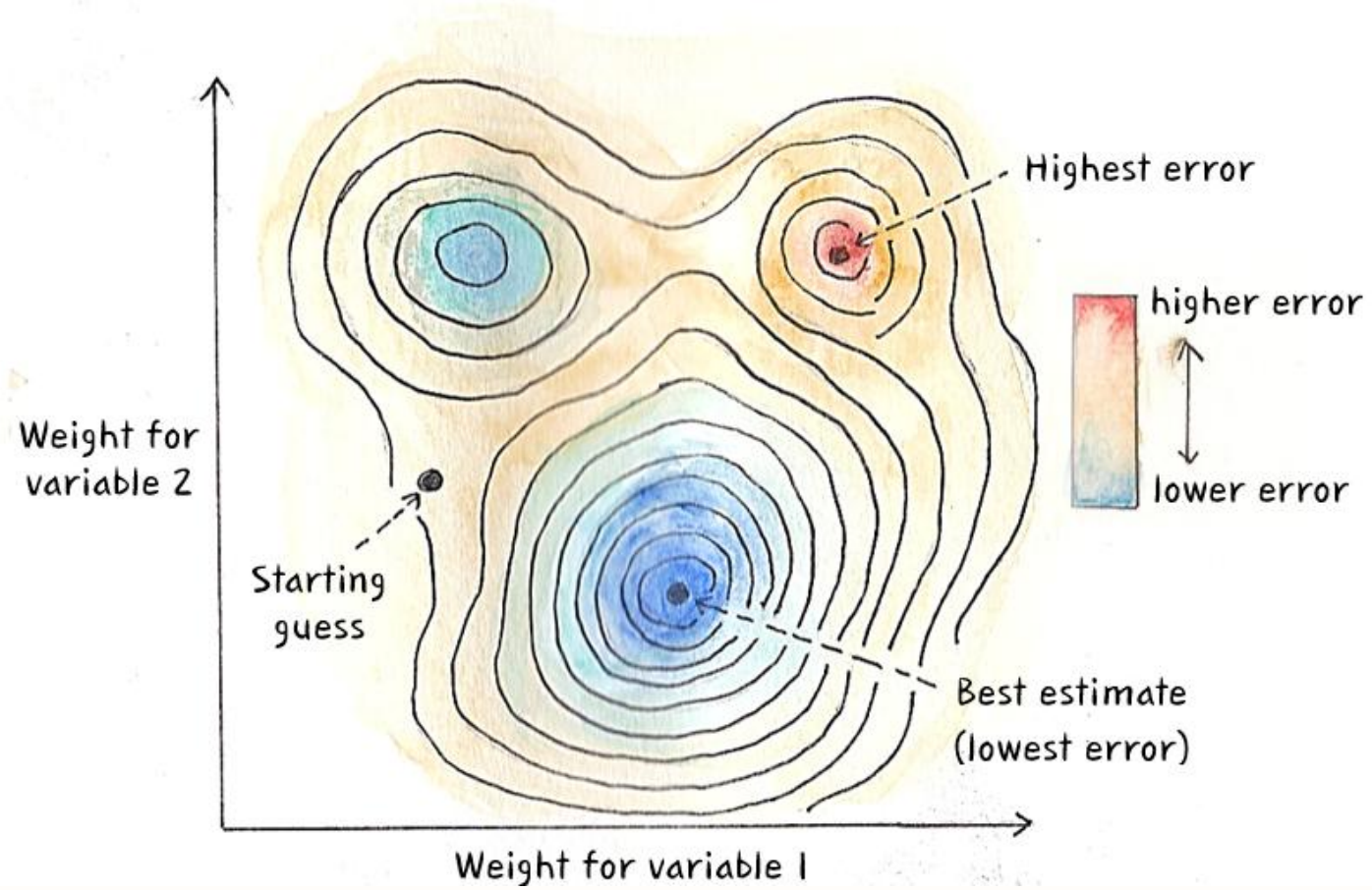
In statistics when we build a model with multiple variables, we are often searching for the set of choices that gives us the lowest error.

“Working out the best answer from incomplete information becomes a search problem. Suppose an outcome is being estimated from several different variables. Each variable is given a weight, which determines how much influence it has on the result…

"With one variable to adjust, the error forms a simple curve, and the lowest point on that curve represents the answer closest to the truth…



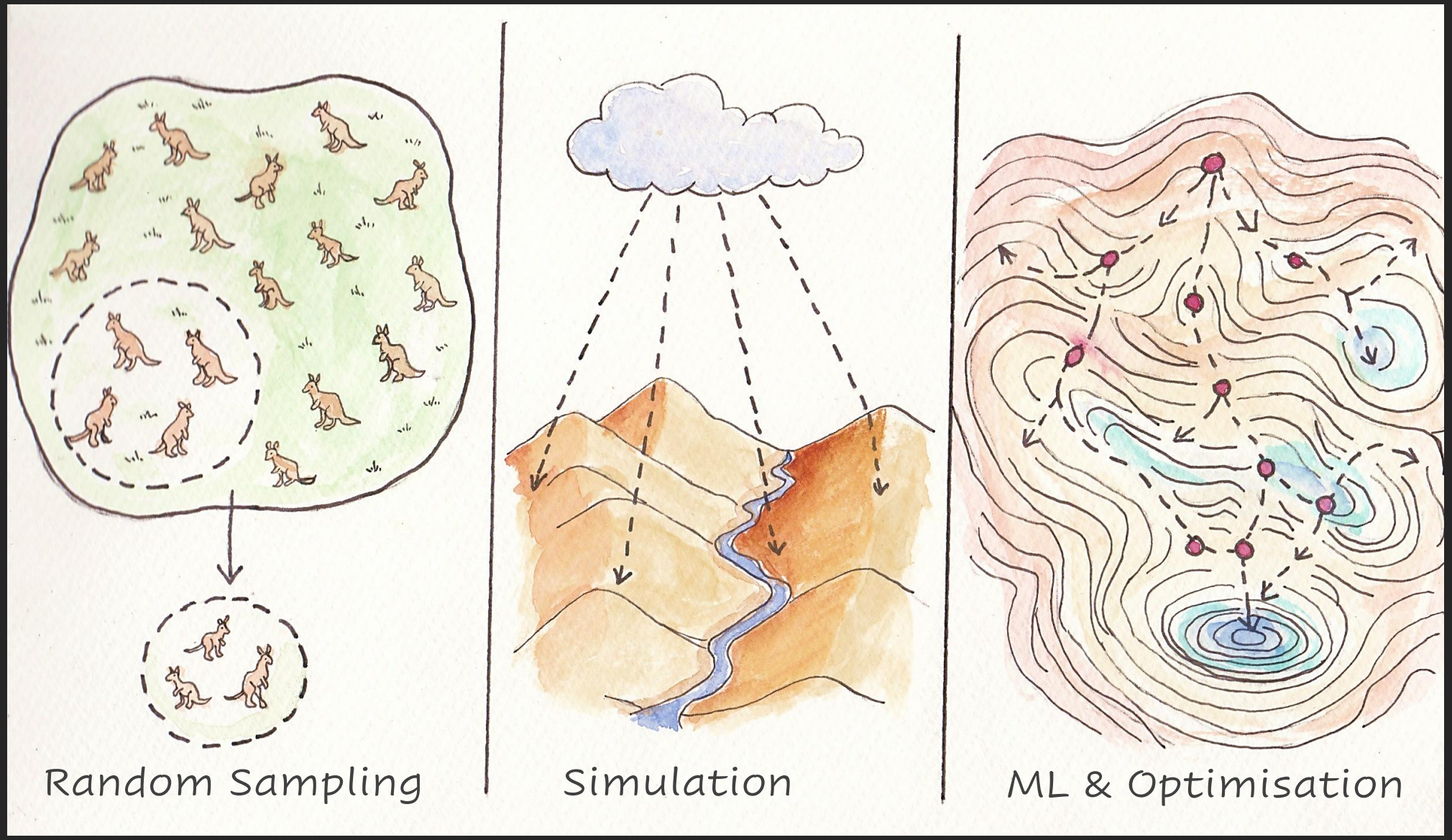
"With two variables, the error forms a surface – something more like a landscape with slopes and valleys. Each position represents a different combination of weights, and the height of the surface shows how wrong those choices are…

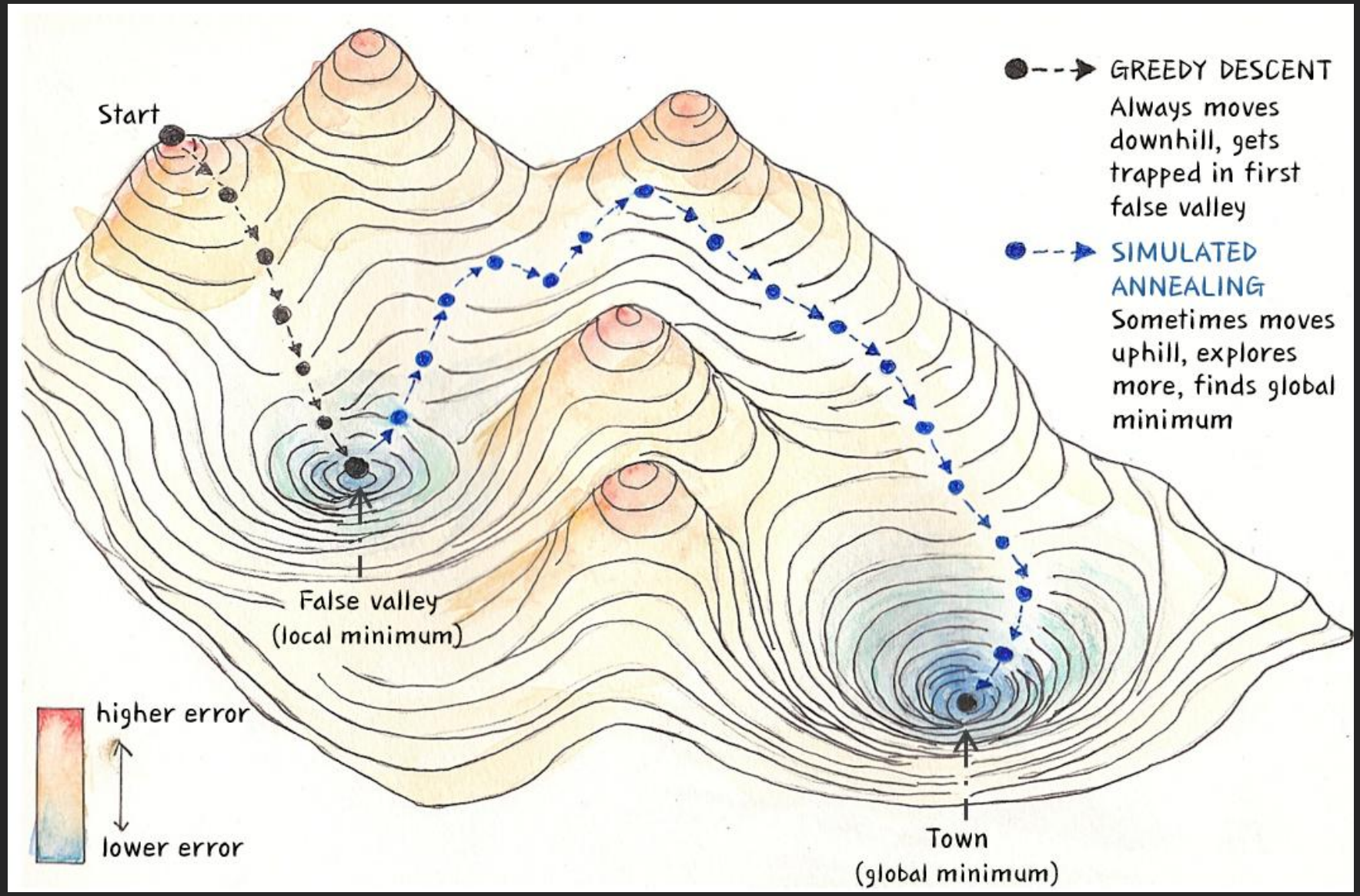


“But real problems rarely stop at two variables. Each additional variable adds another direction the model can move, and the landscape stretches into more dimensions than can be seen. It still exists, but it can no longer be fully drawn or pictured. It simply has…

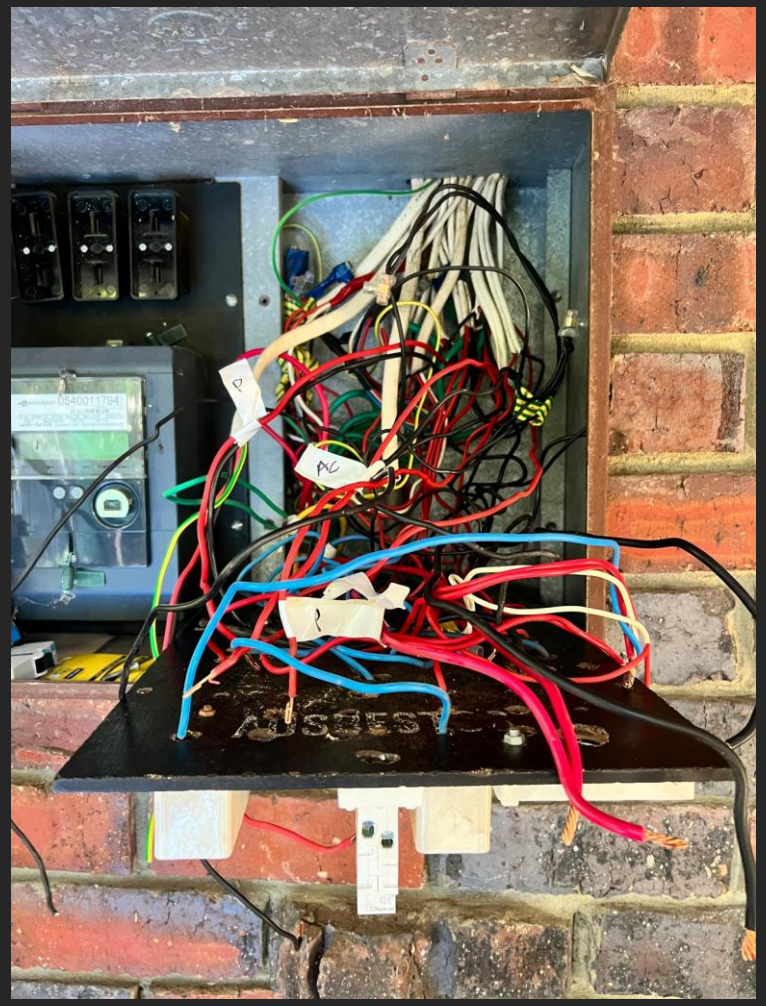


“Her downhill-only strategy had failed because it kept exploiting the best local option.
What she needed now was a strategy that allowed for exploration – a way to avoid getting trapped in the same kind of dead-end low point over and over again…

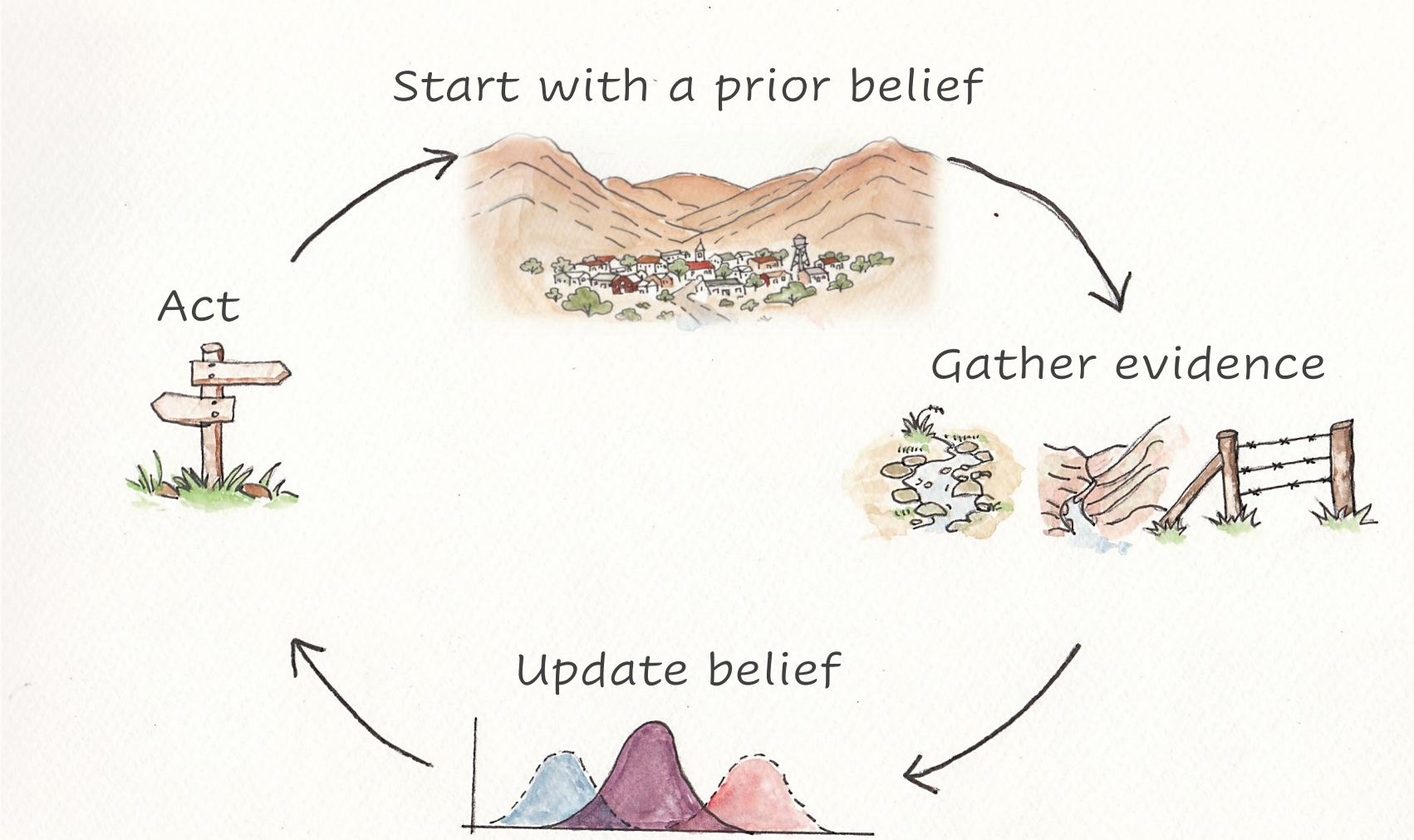

Random Sampling
Simulation
ML & Optimisation

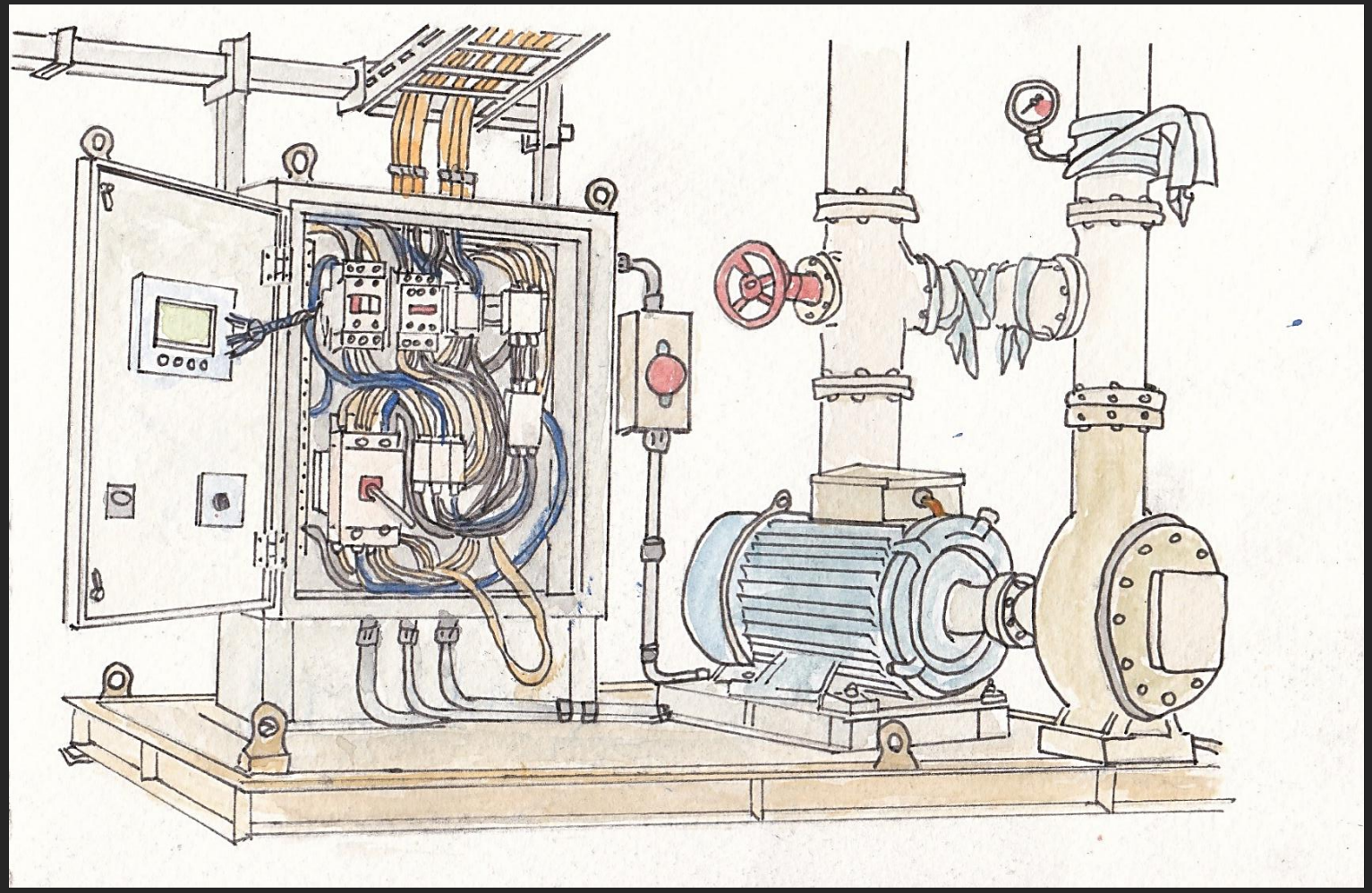

Start
GREEDY DESCENT
Always moves downhill, gets trapped in first false valley
SIMULATED ANNEALING
Sometimes moves uphill, explores more, finds global minimum
False valley
(local minimum)
higher error
lower error
Town
(global minimum)

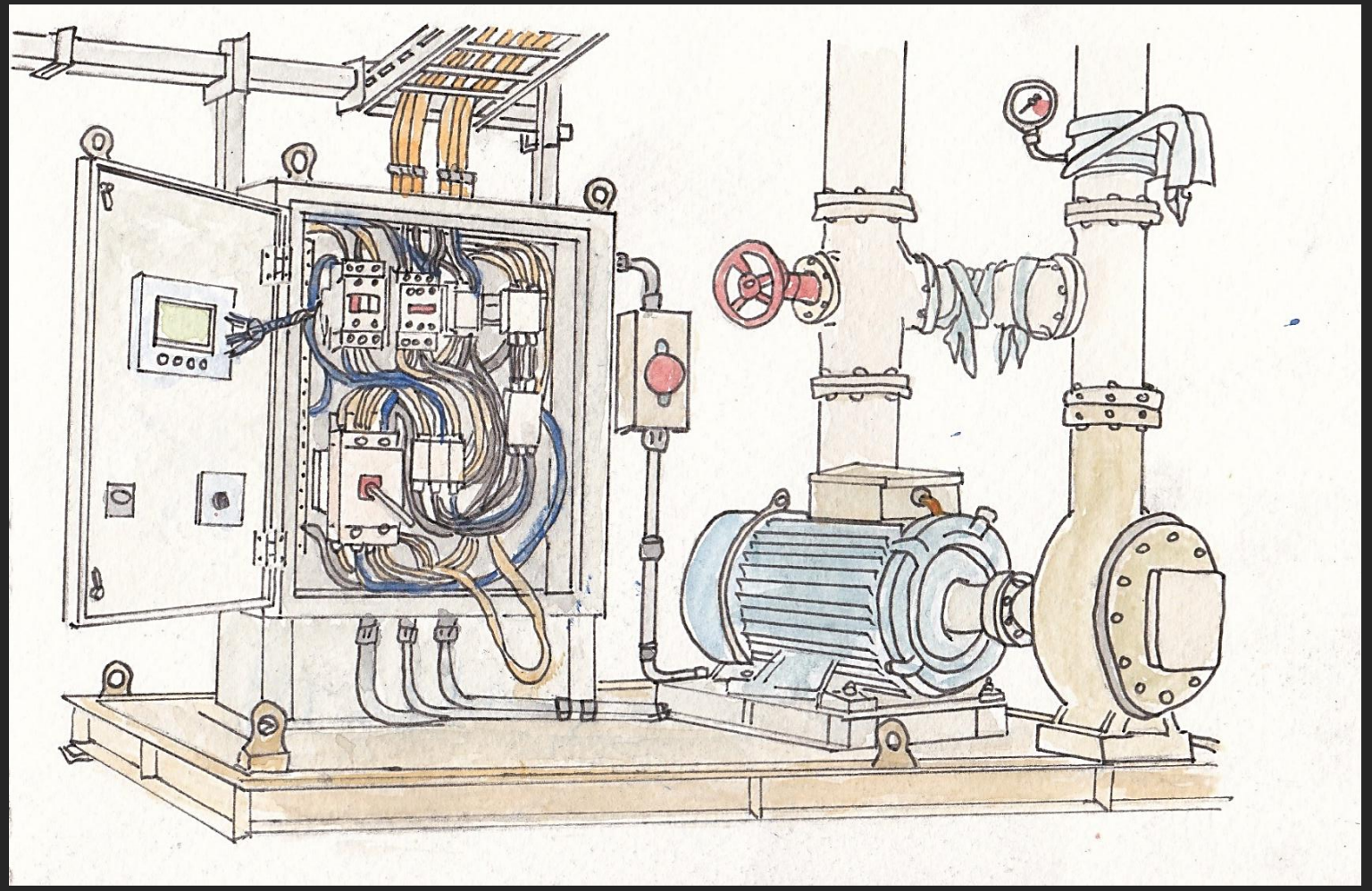


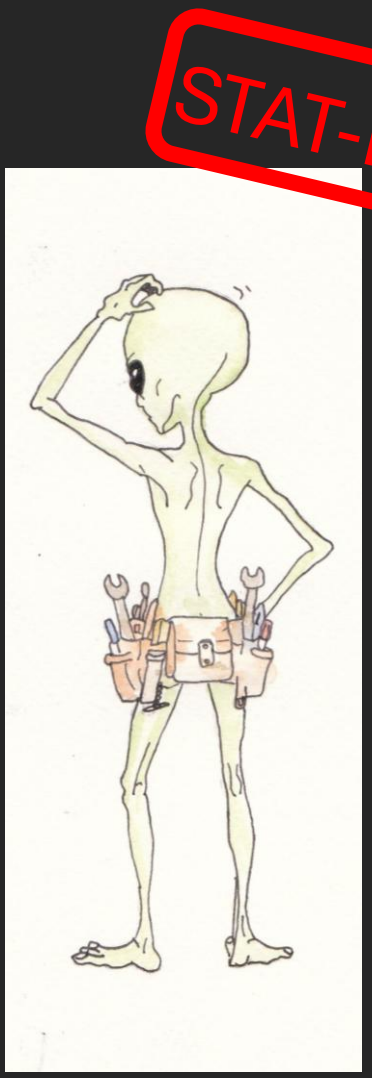


"She started with one prior belief: the town sat in the deepest valley in the range. But knowing that was not the same as knowing how to get there. So as she moved...



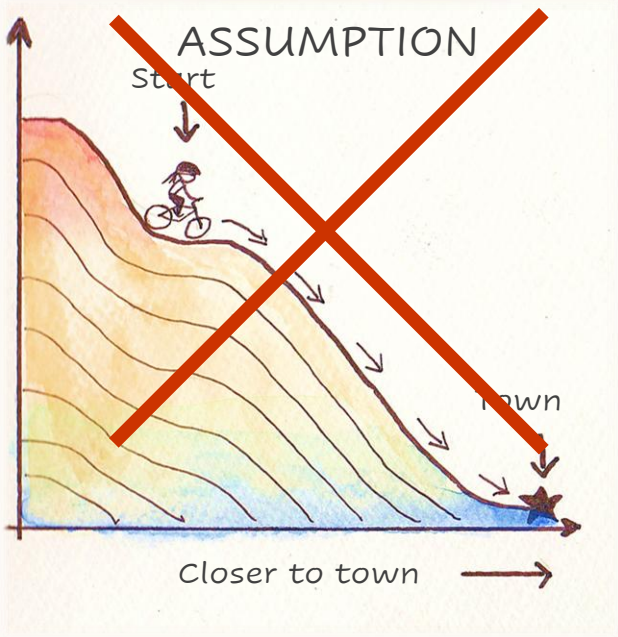




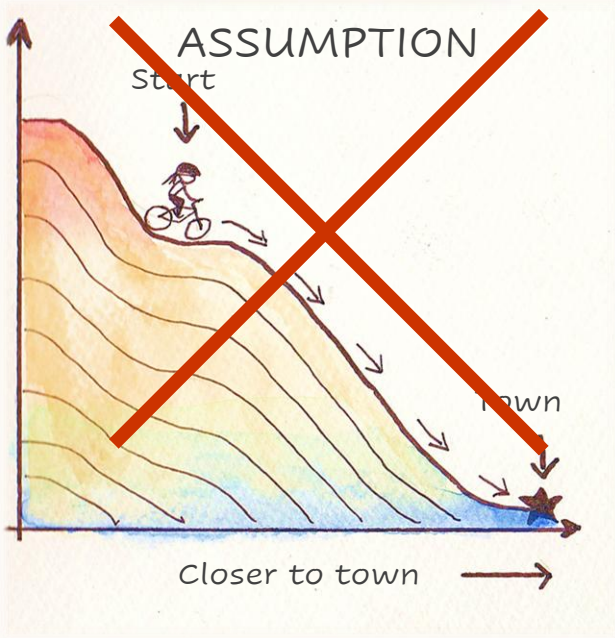

STAT-FI !

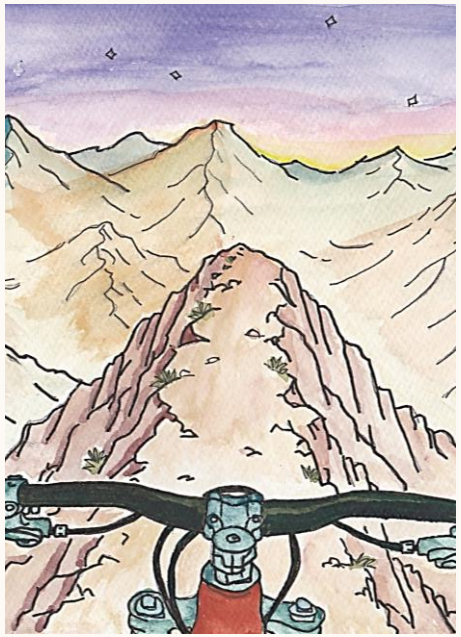
“The mountain no longer felt like her nemesis, just an incomplete dataset. The best route down was not the steepest one beneath her feet. It was the one she could only find by first escaping the local geometry…

the willingness to stop treating the nearest, easiest answer as the best one.

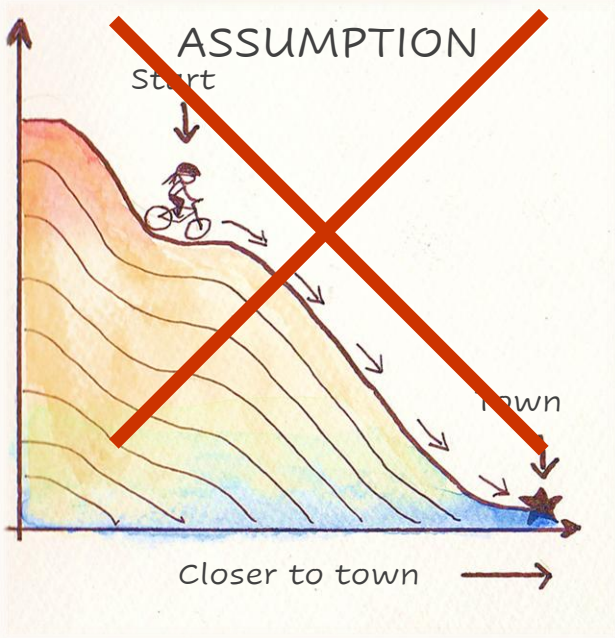


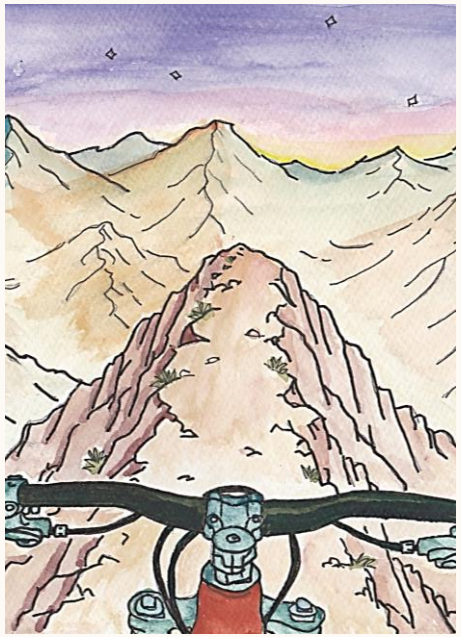
To accept uncertainty.





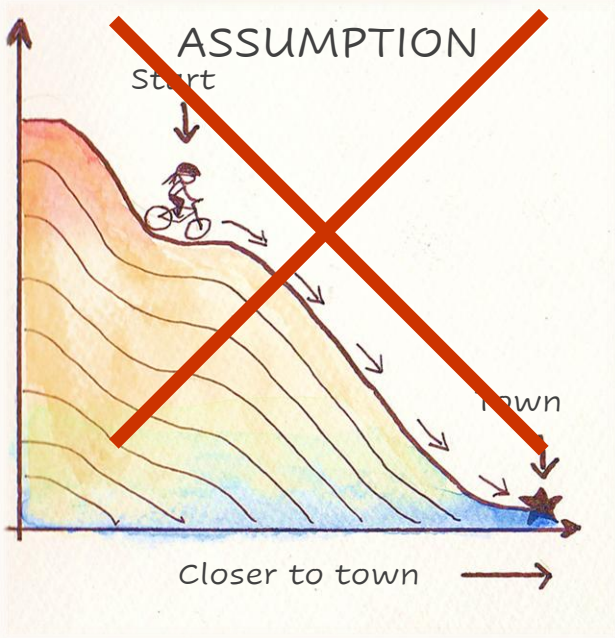
To search through a landscape she could not fully see.

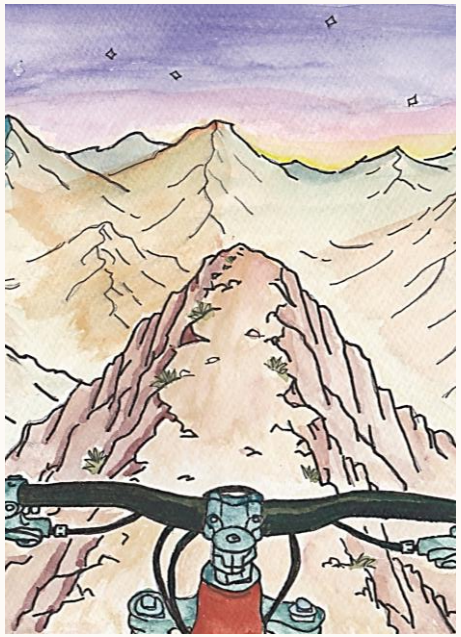




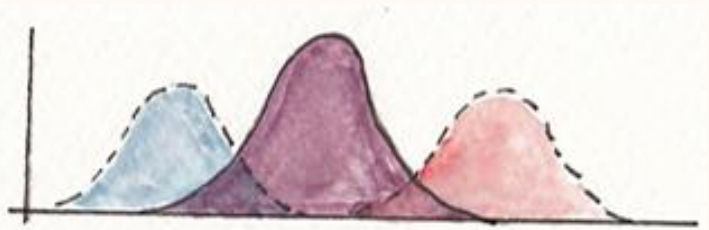


To update beliefs.

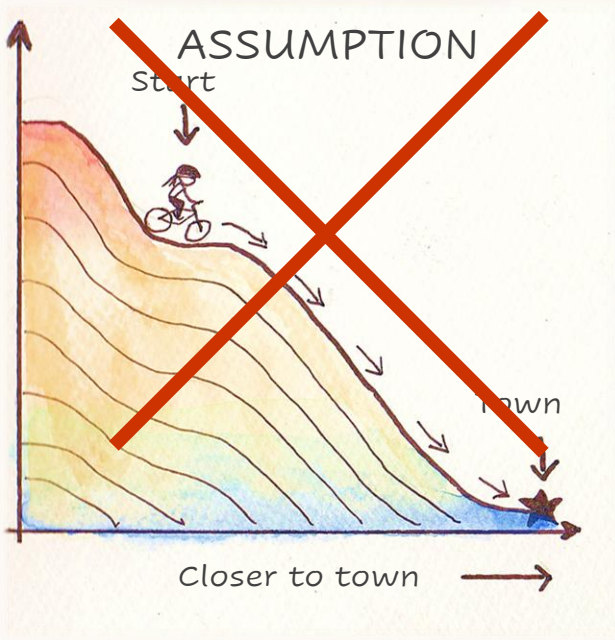


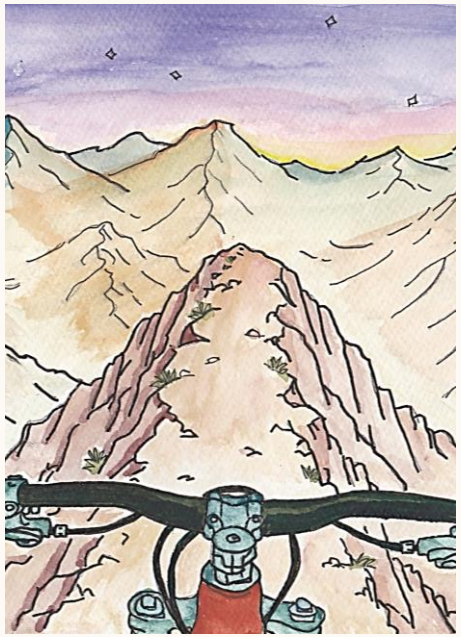




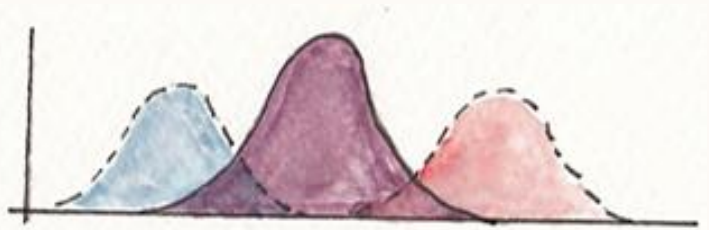


And to allow a little randomness into the search.











Thank you!



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Appendix



